



TANVI MANDAN

24118072

# PERSONALIZED CONTENT DISCOVERY ENGINE

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Netflix Prize Recommendation System



# THE MISSION, IN ONE FRAME

- **Mission:** Build a state-of-the-art engine that learns user taste, predicts ratings, and recommends what they'll love next.
- **Dataset:** Netflix Prize — 100M+ ratings, 480K+ users, 17K+ movies.
- **Winning Model:** Funk SVD — RMSE **0.8643** · MAP@10 **0.7815**.
- **Explainability layer** provides human-readable reasons for every recommendation.
- **Streamlit dashboard** for interactive exploration & demos.

**0.8643**

FUNK SVD · RMSE

**0.7815**

MAP@10

**100M+**

RATINGS ANALYSED

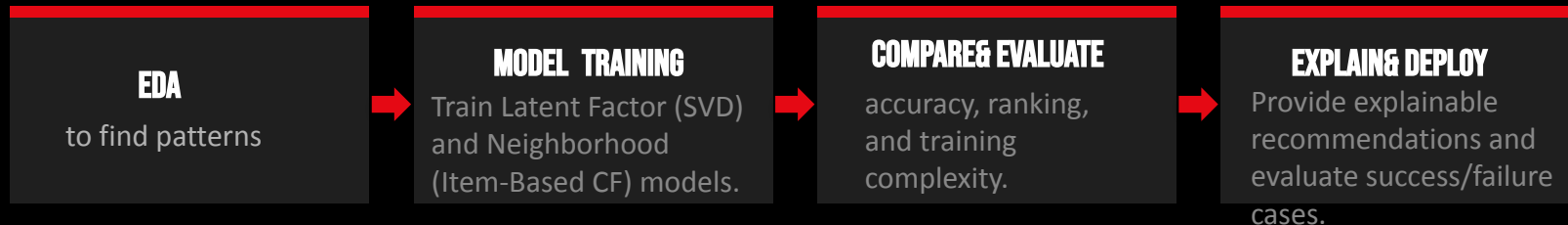
**5×**

FASTER TRAINING

# WHAT WE SET OUT TO SOLVE

- **Objective:** Maximise prediction accuracy (RMSE) and ranking quality (MAP@10).
- **Relevance Threshold:** a movie is 'relevant' if actual rating  $\geq 3.5$  ★.
- **Evaluation across** accuracy, ranking, and training complexity.

## KEY TASKS PIPELINE



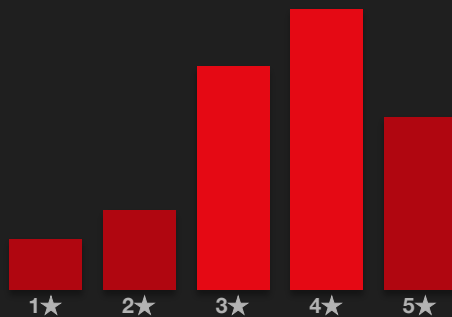
# WHAT THE DATA WHISPERS

## MATRIX SPARSITY

98.82%

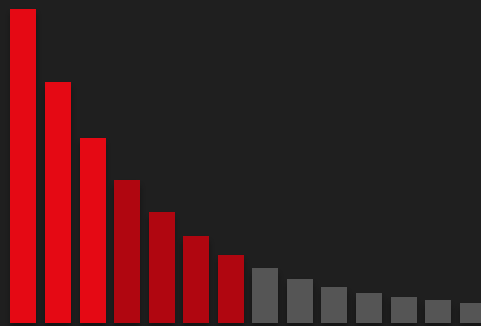
*of the user  $\times$  movie matrix is empty — users rate only a tiny fraction of the catalog.*

## RATING DISTRIBUTION



*Skewed to 3–4 ★ — users rarely log dislikes*

## LONG-TAIL POPULARITY

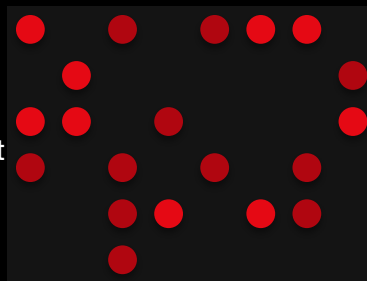


*A few blockbusters dominate; thousands of niche titles stay hidden*

**User activity is long-tailed:** a minority of power-users rate thousands; most rate <50.

# MATRIX FACTORIZATION, CINEMATIC EDITION

- **Approach:** Factor the sparse  $R$  into user ( $P$ ) and item ( $Q$ ) embeddings of size  $K$ .
- **Optimisation:** Minimise MSE with L2 regularisation using Adam in PyTorch.
- **Why it wins:** Highly scalable, robust to sparsity, learns latent taste.
- **Captures:** global mean  $\mu$ , user bias  $b_u$ , item bias  $b_i$  + latent interaction  $P_u \cdot Q_i^T$ .

 $R$  (users  $\times$  items) $P$  (users  $\times$   $K$ ) $Q^T$  ( $K \times$  items) $\approx$  $\times$ 

## PREDICTION FORMULA

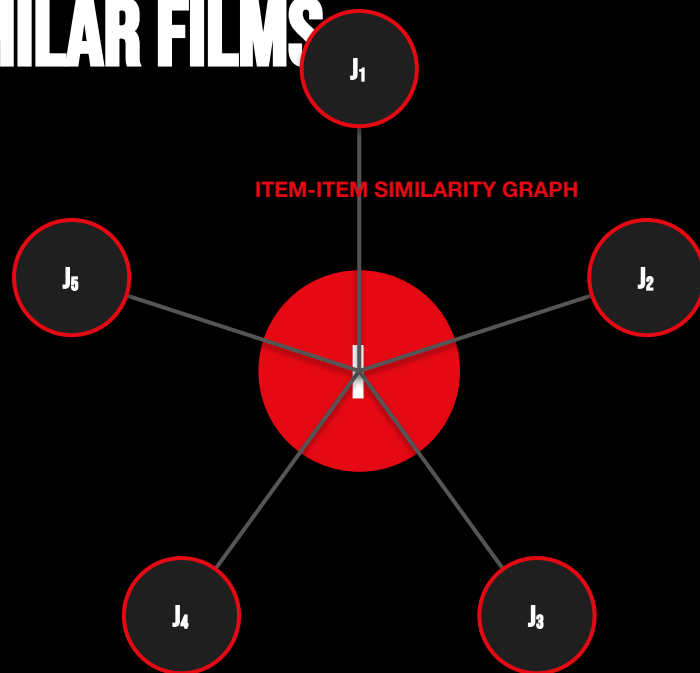
$$\hat{r}_{ui} = \mu + b_u + b_i + P_u \cdot Q_i^T$$

# NEIGHBOURHOOD METHOD — FIND SIMILAR FILMS

- **Approach:** Memory based compute cosine similarity between items from rating vectors.
- **Highly explainable** (transparent neighbour weights), but heavy at scale.
- **Trade off:** struggles with extreme sparsity; no latent generalisation.
- **Cost:** computes a full  $M \times M$  similarity matrix.

## INTERPOLATION FORMULA

$$\hat{r}_{ui} = \sum w_i \cdot r_{ui} / \sum |w_i|$$



Predict  $r_{ui}$  from its top-K nearest items the user already rated.



# FUNK SVD TAKES THE CROWN

| METRIC           | FUNK SVD | ITEM-BASED CF |
|------------------|----------|---------------|
| RMSE (Accuracy)  | 0.8643   | 0.9412        |
| MAE              | 0.6720   | 0.7285        |
| MAP@10 (Ranking) | 0.7815   | 0.7104        |
| Training Time    | 2.5 s    | 12.3 s        |

## TOP TAKEAWAYS

**Funk SVD wins on every metric.**

Latent representations generalise better across sparse users and items.

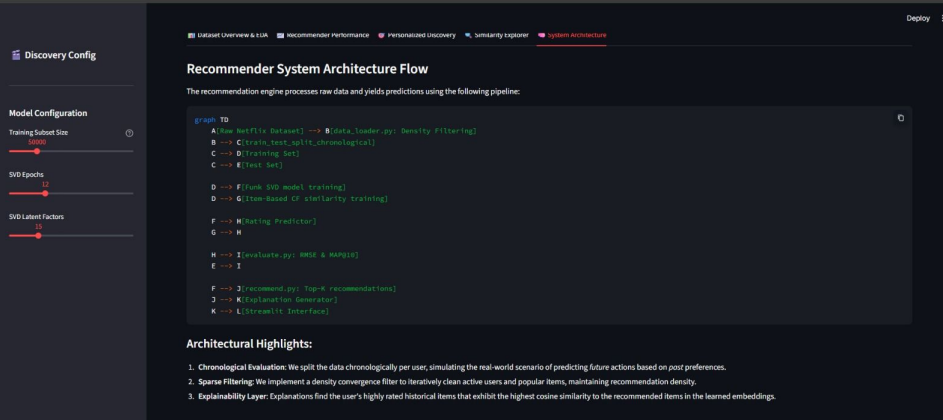
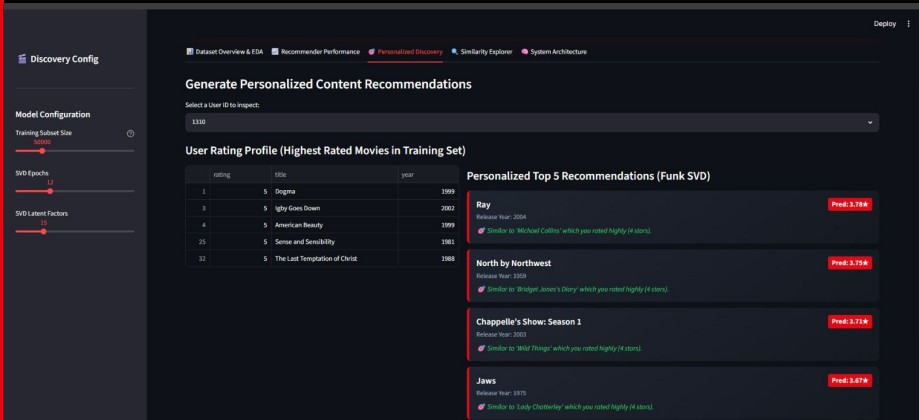
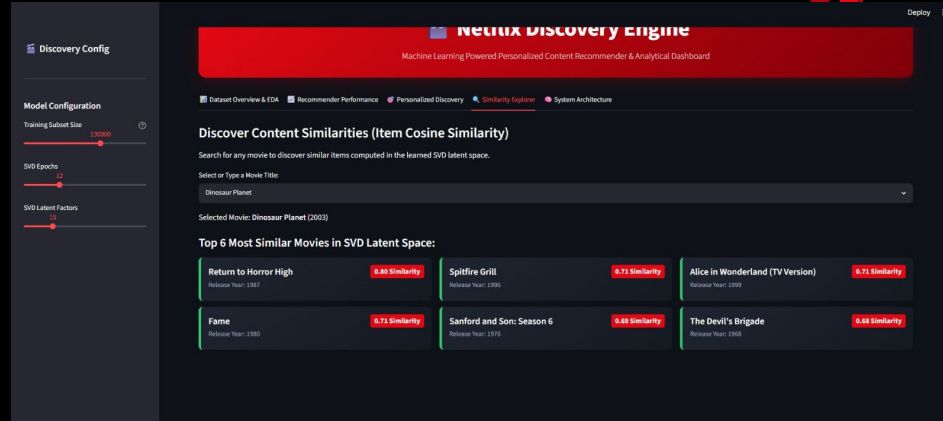
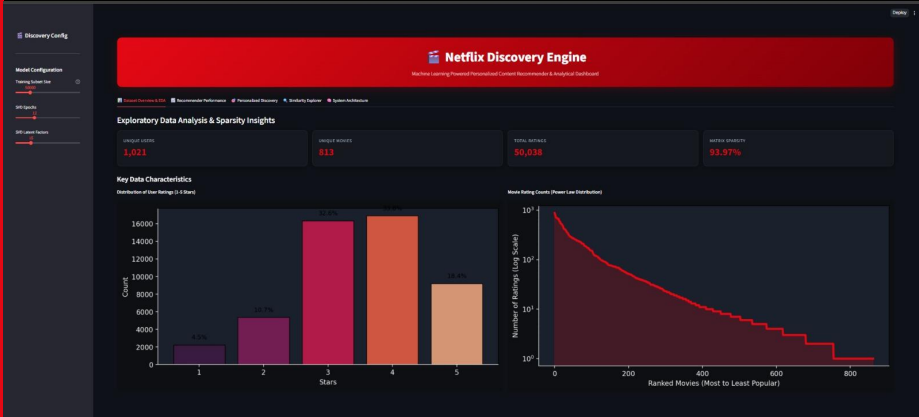
**≈ 5× FASTER TRAINING**



EPISODE 07 LOCAL INTERACTIVE PROTOTYPE

# PROTOTYPE





# BECAUSE YOU WATCHED...

TOP MATCH

★ 4.65

## THE GODFATHER

WHY YOU'LL LIKE IT

*Similar to "Scarface" which you rated 5.0 ★ in the past.*

TOP MATCH

★ 4.48

## LORD OF THE RINGS

WHY YOU'LL LIKE IT

*Similar to "Harry Potter" which you rated 4.5 ★ in the past.*

**Insight** — transparent reasons build user trust and lift engagement.



# COMING NEXT SEASON

## COLD START STRATEGY

- **Ask new users** to pick 3–5 favourite genres or films at signup.
- **New items** represented via content features (genres, directors) mapped into latent space.
- **Bootstrap quickly** with content-based fallback until enough signals arrive.

## SYSTEM ROADMAP

### HYBRID — LightFM

Blend metadata (genres, actors) with collaborative ratings.

### DEEP LEARNING — NCF

Capture non-linear user Item interactions with neural collaborative filtering.

### SEQUENCE — SASRec

Model temporal viewing journeys for next-up recommendations.

► THE STORY CONTINUES · THANK YOU